

A125 – The φ Skeleton: from the Weights to the Mass Ratios

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October 2026

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.1 Purpose

A120 shows that the fivefold rotation R_5 of the icosahedron redistributes the three Z_3 modes with the weights

$$p_0 = \frac{2}{9}, \quad p_1 = \frac{2+3\varphi}{9}, \quad p_2 = \frac{5-3\varphi}{9},$$

and leaves one edge open: how these weights are connected with the mass ratios of the charged leptons. This document closes that edge as far as the corpus closes it today. It works out two routes, both of which give pure ratios, without an anchor and without K_{frak} : the Koide formula from a single matrix element of R_5 , and the φ skeleton of the weights themselves. On the second route

$$\frac{m_\tau}{m_e} = \left(74 + \frac{1}{45}\right)\varphi^8 = 3477.469$$

agrees with the measured value to 3×10^{-5} , within the measurement uncertainty. The document states just as clearly what the route does not achieve: the factors 74 and $1/45$ are observed, not derived, and for m_μ/m_e there is no φ route at this level.

[STIPULATION] As in A120, the embedding of the C_3 structure in the icosahedral group A_5 is presupposed.

.2 Why the icosahedron: the three and the five

[B] FFGFT is a Z_3 theory; the golden ratio is the number of the five. The rotation group A_5 of the icosahedron is the smallest finite group in which a threefold and a fivefold rotation occur together without commuting. This non-commuting produces the mixing of the Z_3 modes under R_5 that A120 computes.

[B] The two symmetries enter at different levels. The three acts as a lattice operation on the compact carrier: the triality of the D_4 lattice is the Z_3 of the theory. The lattice knows no five, since $|\text{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$ and $5 \nmid 1152$; no icosahedron acts on a D_4 lattice. The five enters only in the unrolled space $\mathbb{R}^3 \times S^1$ (A090), and it does so as a phase: the 72° rotation has the characteristic polynomial $(x-1)(x^2 - (\varphi-1)x + 1)$ with trace φ . Modulo 3, φ lies in $\text{GF}(9)$ and is primitive there ($\varphi^4 = -1$, order 8); the eigenvalues ζ_5 lie in $\text{GF}(81)$, because $5 \mid 80 = |\text{GF}(81)^*|$.

[B] The distribution over the two levels is forced, not chosen. In characteristic 3 one has $x^3 - 1 = (x-1)^3$; no $\text{GF}(3^k)$ contains a primitive third root of unity. The three therefore cannot be a phase and must be lattice; the five cannot be lattice and must be phase. Three and five do not meet in the geometry but in the field $\text{GF}(81)$; the extension $\text{GF}(9) \rightarrow \text{GF}(81)$ is the doubling $6 = 3 \oplus 3'$.

.3 One matrix element carries the Koide formula

[B] The weight $p_0 = 2/9$ and the Koide angle $\theta = 2/9$ (A110, A120) are the same quotient $2/3^2$: in the numerator the number of non-trivial Z_3 modes, in the denominator the square of the group order. For the matrix element $A = \langle v_0 | R_5 v_e \rangle$ one has $|A| = \sqrt{2}/3$; the modulus gives the Koide amplitude, the squared modulus the phase:

$$a_k = 1 + 3|A| \cos(|A|^2 + \frac{2\pi k}{3}), \quad 3|A| = \sqrt{2}, \quad |A|^2 = \frac{2}{9}.$$

The same number appears once as the modulus and once as the squared modulus of the same matrix element. The Koide formula thus contains no free parameter any more; the mass amplitudes follow from the A_5 geometry.

[K] The ratios $(a_0/a_1)^2$ and $(a_2/a_1)^2$ give

Ratio	Formula	Value	Measured	Deviation
m_τ/m_e	$(a_0/a_1)^2$	3477.473	3477.37 ± 0.18	$+3.1 \times 10^{-5} \text{ (0.6}\sigma\text{)}$
m_μ/m_e	$(a_2/a_1)^2$	206.77032	$206.7682827(46)$	$+9.8 \times 10^{-6} \text{ (442}\sigma\text{)}$

For m_τ/m_e the value lies within the measurement uncertainty of m_τ . For m_μ/m_e the exact pair $(\sqrt{2}, 2/9)$ is excluded: at fixed amplitude the ratio depends only on θ , and the deviation exceeds the uncertainty by more than two orders of magnitude. This is not an objection to the origin of $2/9$ but a limit of the exact pair.

[B] Since $2/9 \notin \pi\mathbb{Q}$, $\cos(2/9)$ is transcendental. Exact algebraic mass amplitudes are therefore excluded in principle on this route; any algebraic form can only approximate the Koide value.

.4 The φ skeleton of the weights

[B] The two weights containing φ stand in exact ratios to each other and to the rational weight:

$$\frac{p_1}{p_2} = \varphi^8, \quad \frac{p_0}{p_2} = 2\varphi^4, \quad \frac{p_1}{p_0} = \frac{\varphi^4}{2}, \quad 3\sqrt{p_j} \in \{\sqrt{2}, \varphi^2, \varphi^{-2}\}.$$

These are pure numbers from the geometry, without a cosine and without a scale. The Koide amplitude $\sqrt{2} = 3\sqrt{p_0}$ stands in the same row as φ^2 and φ^{-2} .

.5 From the skeleton to the mass ratios

[K] The mass ratios carry this skeleton. The most precise hit is

$$\frac{m_\tau}{m_e} = \left(74 + \frac{1}{45}\right)\varphi^8 = \frac{3331}{45}\varphi^8 = 3477.469,$$

that is $+3.0 \times 10^{-5}$ or 0.6σ against the measured value 3477.37 ± 0.18 (PDG 2024). The overview:

Ratio	Formula	Value	against measured value
m_τ/m_e	$74\varphi^8$	3476.42	-2.7×10^{-4} (-5.3σ)
m_τ/m_e	$(74 + \frac{1}{45})\varphi^8$	3477.469	$+3.0 \times 10^{-5}$ (0.6σ)
m_τ/m_e	$74\varphi^8(1 + \frac{27}{125}\xi)$	3477.468	$+2.9 \times 10^{-5}$ (0.6σ)
m_τ/m_e	Koide, $\theta = 2/9$	3477.473	$+3.1 \times 10^{-5}$ (0.6σ)
m_μ/m_e	$30\varphi^4 = 15 p_0/p_2$	205.62	-0.55%

[K] On the φ route, m_τ/m_e is therefore as precise as on the Koide route, as a pure ratio in natural units. The route is algebraic: the φ form approximates the transcendental Koide value, and the second layer ($1/45$ or $\frac{27}{125}\xi$) carries the remainder. Against the Koide value itself, $(74 + \frac{1}{45})\varphi^8$ is off by only 1.2×10^{-6} .

[S] The factors are observed, not derived. $74 = 2 \cdot 37$ is the same number as in the numerator of the rotation number $74/75 = 1 - 100\xi$ of the recursion (A050) and in the value of K_{frak} (A040). The second layer has the same form as the Galois form $1/\alpha = 3700/27 = 137 + \frac{1}{27}$ (A130): an integer plus a unit fraction. The 37 appears only in numerator parts ($74 = 2 \cdot 37$, $3700 = 100 \cdot 37$), in no denominator ($45 = 3^2 \cdot 5$, $27 = 3^3$); a common algebraic source is therefore not established.

[B] The 37 demonstrably does not come from the icosahedron. In the phase channel it is the root of unity ζ_{37} , which lives in $\text{GF}(3^{18})$, because the order of 3 modulo 37 is 18. The icosahedral phase ζ_5 lives in $\text{GF}(81) = \text{GF}(3^4)$, and $4 \nmid 18$: $\text{GF}(81)$ is not a subfield of $\text{GF}(3^{18})$. The two fields share only $\text{GF}(9)$, that is, φ .

.6 Two readings of the same numbers

[B] The ladder of A100 and the Koide route give the same numbers, but they can be read in two directions. In reading A the ladder $r_k \xi^{p_k}$ is fundamental, and the step from the per-cent level to the 10^{-5} level is a measured correction. In reading B the Koide formula from the icosahedron is fundamental; the ladder is then a rational approximation to the transcendental amplitudes a_k^2 , and the generation-dependent residuals are the distance between cosine and power law, computed rather than measured **[K]**. They differ for the three leptons because the three cosine values differ. Which reading carries the weight is not decided by the calculation **[S]**.

.7 What the route does not achieve

[S] For m_μ/m_e there is no φ route at the level of m_τ/m_e : $30\varphi^4$ is off by 0.55% , the ladder of A100 by 0.52% , and the exact Koide pair is excluded (above).

[K] The φ skeleton is not the earlier φ mass ladder with $m \propto \varphi^{\text{generation}}$ and $m_\mu/m_e \approx \varphi^{10}$. That ladder is refuted: $\varphi^{10} = 122.99$, not 206.77 .

.8 Verification Script

- Weights p_j from R_5 (exactly against $2/9$, $(2+3\varphi)/9$, $(5-3\varphi)/9$, $|A| = \sqrt{2}/3$, the φ skeleton, the Koide ratios against PDG 2024 and CODATA, the table of φ forms, the field statements ($5 \nmid 1152$, $\varphi^4 = -1$ in $\text{GF}(9)$, order of 3 modulo 37, $4 \nmid 18$) and the distinction from φ^{10} :
python/A_Serie_Skripte/a125_phi_skelett.py

.9 Symbol Glossary

Symbols used throughout the A series are explained in A010. Specific to this document:

Symbol	Meaning
φ	golden ratio $(1 + \sqrt{5})/2$
A_5, R_5	rotation group of the icosahedron; fivefold rotation by 72°
v_0, v_e	trivial and non-trivial (electron) mode of the Z_3 circulant
p_j	redistribution weights $ \langle v_j R_5 v_e \rangle ^2$
A	matrix element $\langle v_0 R_5 v_e \rangle$, $ A = \sqrt{2}/3$
a_k	Koide amplitudes $1 + \sqrt{2} \cos(\theta + 2\pi k/3)$
ζ_n	primitive n -th root of unity

.10 Open Questions

First, the factors 74 and $1/45$. They do not come from the weights themselves and demonstrably not from the icosahedron. Whether they follow from the Galois hierarchy $\text{GF}(3^n)$, in which K_{frak} and $3700/27$ also stand, is open.

Second, the common root. The icosahedron supplies phases in $\text{GF}(81)$, number theory supplies orders in $\text{GF}(3^{18})$; both hit the lepton masses, but they touch only in φ . Whether there is a structure in which both have the same root, or whether they are two independent approaches, is the open edge.

Third, m_μ/m_e . For this ratio an algebraic route at the level of 10^{-5} is missing.

Fourth, the embedding $C_3 \subset A_5$. As in A120, it is presupposed.

.11 Supersedes

This document incorporates: Doc. 367 (one matrix element carries Koide), Doc. 368 (φ skeleton and Galois factors), Doc. 369 (routes 2 and 4 of the inventory), Doc. 370 (why the icosahedron), Doc. 364 (as far as A_5 and D_4 are concerned). Incorporated: R121. Delimited: Docs. 172, 281 (refuted φ mass ladder).

.12 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.